

Machine Learning Project

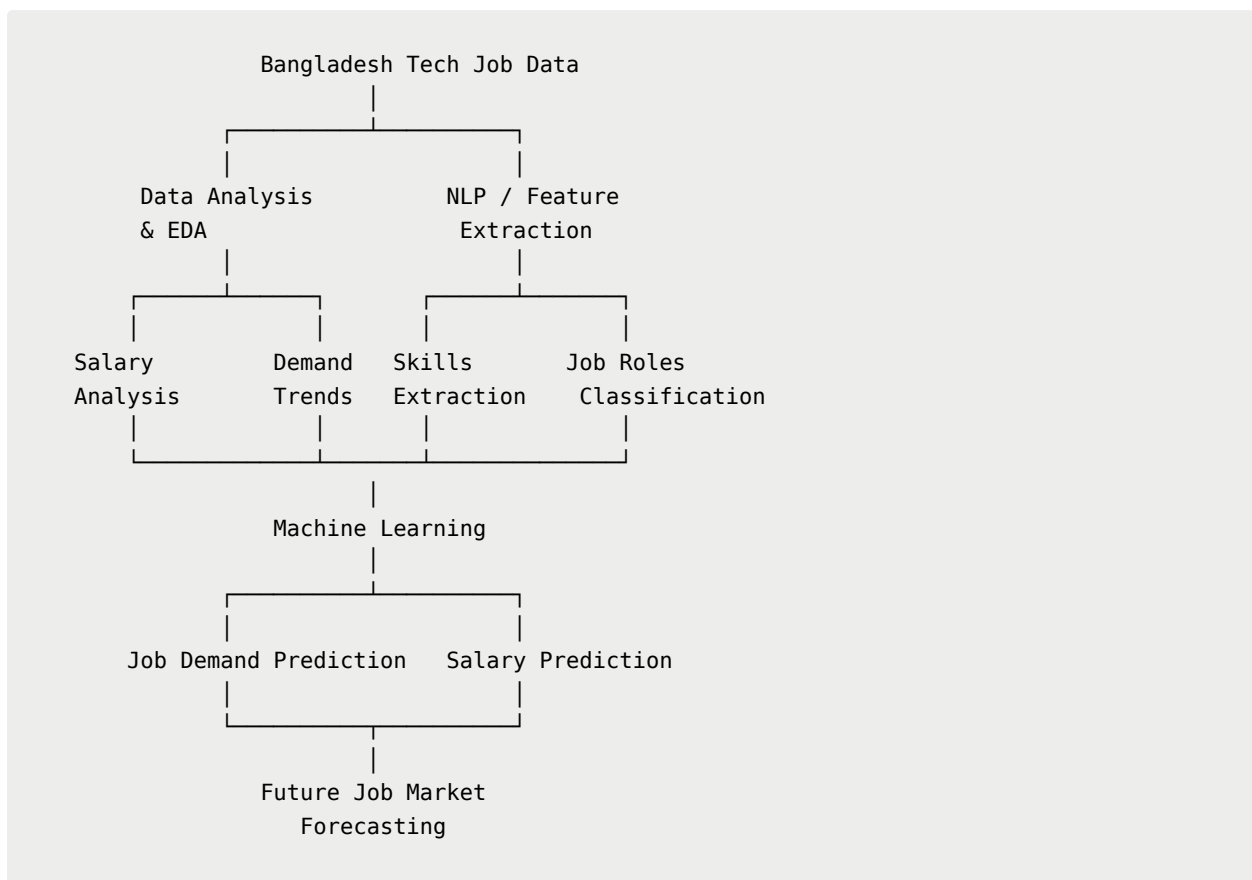
Project title

TechCarrierBD: Machine Learning-Based Analysis and Forecasting of Bangladesh's Technology Job Market

A more academic title for the IEEE paper:

Machine Learning-Based Analysis of In-Demand Skills, Salary Trends, and Job Demand Forecasting in Bangladesh's Technology Sector

Your system can have **three analytical components**:



However, **you don't need to make every box a separate ML problem.**

1. First: Define the research questions

Your project should answer 4–5 specific questions.

RQ1 — Current demand

Which technology job roles and skills are most frequently demanded in Bangladesh?

Example:

Software Engineer	32%
Web Developer	21%
Data Analyst	15%
Network Engineer	11%
...	

And:

Python
JavaScript
SQL
React
Java
AWS
Docker
...

This is primarily **EDA / NLP**, not necessarily your main ML task.

RQ2 — Salary

How do advertised salaries vary according to job role, experience, location, and required skills?

For example:

Role	Experience	Median Salary
Software Engineer	Entry	X
Software Engineer	Mid	X
Software Engineer	Senior	X

Data Analyst	Entry	X
...		

Again, this can be statistical analysis rather than your primary ML task.

RQ3 — Job demand prediction

This should become your **main supervised ML problem**.

For example:

Given job characteristics, can we predict the job category/role?

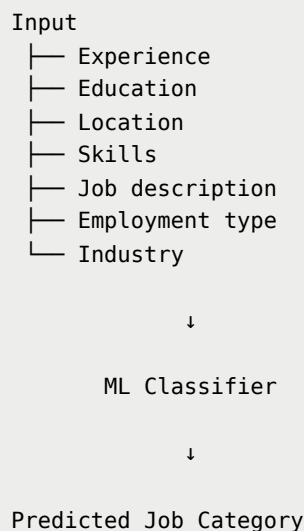
or

Given job-posting characteristics, can we predict whether a posting belongs to a high-demand or low-demand category?

I would recommend **classification** because it fits your professor's requirements very naturally.

Based on job demand, it will help us learn which skills a user should learn. Which framework is in highest demand?

For example:



Possible target:

```
Software Development
Data / AI
Cybersecurity
Networking / Cloud
IT Support
UI/UX
Other
```

Then you can directly compare:

- Logistic Regression → baseline
- Decision Tree → baseline/alternative
- Random Forest → advanced
- XGBoost → advanced
- SVM → advanced candidate

That satisfies your professor's **baseline vs. advanced modeling** requirement.

2. Add a second ML problem if your dataset supports it

This would make the project considerably stronger.

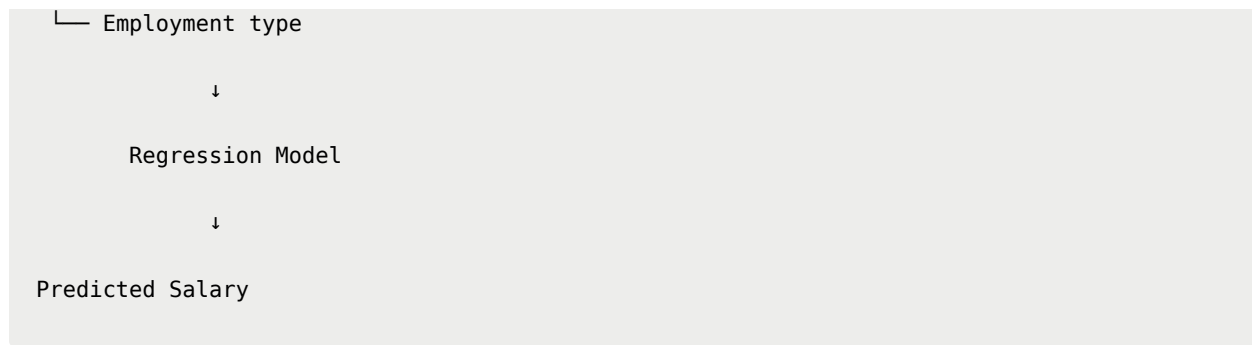
Salary prediction

If enough job advertisements contain salary information:

Can we predict the salary range of a technology job based on its role, experience, location, skills, and other characteristics?

This becomes a **regression problem**.

```
Features
├─ Job role
├─ Experience
├─ Location
├─ Skills
└─ Education
```



Models:

Baseline

- Mean/Median predictor
- Linear Regression

Advanced

- Random Forest Regressor
- XGBoost Regressor
- Gradient Boosting

Evaluation:

- MAE
- RMSE
- R^2

But there is an important condition:

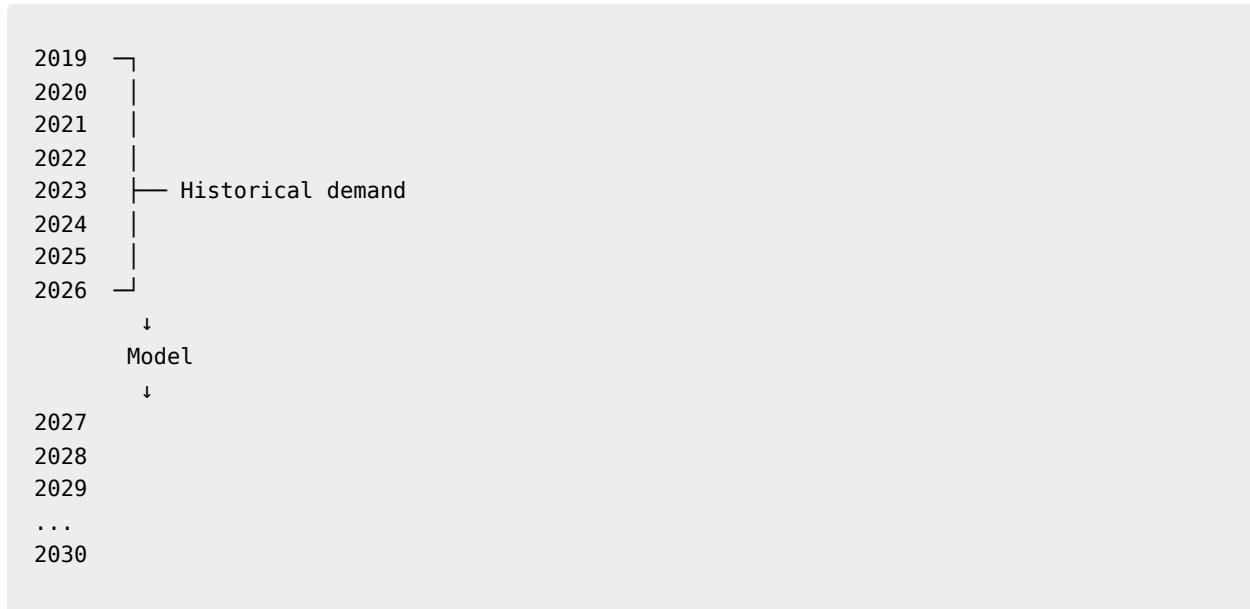
If salary is missing from a large proportion of your data, don't force this problem.

You can still provide salary **descriptive analysis** without making salary prediction your ML task.

3. Future forecasting should be a separate time-series experiment

"We forecast the future demand trend of selected technology occupations using historical job-posting data."

For example:



You could forecast:

- Software Developer demand
- Data Analyst demand
- AI/ML demand
- Cloud/DevOps demand
- Cybersecurity demand

Important

Your professor specifically wants **recent data**, so don't build the project around the 2016–2021 dataset from the existing Bangladesh paper.

That paper is useful as **related work**, not necessarily as your primary dataset.

The 2022 ICT study used **31,526 Bdjobs postings from 2016–2021** and analyzed job titles and skills using NLP. ([MDPI](#))

Another Bangladesh study used Bdjobs data, NLP, and ARIMA to predict job-skill demand. ([DU Journal of Business Studies](#))

That gives you an excellent foundation for your literature review.

4. Your biggest task: Build a recent dataset

This should be your **Phase 1 priority**.

Ideally collect something like:

Field	Example
Job Title	Software Engineer
Company	Company A
Location	Dhaka
Posted Date	2026-08-15
Experience	2-4 years
Education	BSc
Salary Min	40,000
Salary Max	70,000
Employment Type	Full-time
Job Description	...
Requirements	...
Source	Portal name
URL/ID	...

And then create derived fields:

```
job_category
experience_level
salary_midpoint
skills
skill_count
location_category
year
month
```

5. Don't collect only one month's data

This is **very important** for your project.

If you collect:

```
August 2026
↓
5,000 jobs
```

you can do:

- EDA
- skill analysis
- salary analysis
- classification

But you **cannot convincingly perform long-term forecasting**.

For forecasting, you need:

```
Month 1
Month 2
Month 3
...
Month 24
Month 36
...
```

Ideally:

Minimum

2–3 years of monthly observations

Better

3–5 years

The historical Bangladesh ICT study is useful partly because it had a multi-year time series. ([MDPI](#))

6. Your dataset should ideally contain 10,000+ postings

I'd target something like:

| **10,000–50,000 technology job postings**

More important than the raw number is:

quality + timestamps + consistent fields + enough salary observations.

For example:

```
30,000 postings
    ↓
10,000 duplicates/low-quality
    ↓
20,000 usable postings
    ↓
15,000 with useful descriptions
    ↓
6,000 with salary information
```

That's completely acceptable if you document the filtering process.

7. Use existing research to design your methodology

This is where your professor's "reference paper" suggestion becomes useful.

Your literature review can contain something like:

Paper 1

An Exploratory Study of Online Job Portal Data of the ICT Sector in Bangladesh

This study analyzed 31,526 ICT job postings from Bdjobs from 2016–2021 and used NLP to group job titles and skills. It found changes in demand for specific ICT roles and skills over time. ([MDPI](#))

Paper 2

Predicting Job Skills in Demand: A Big Data Approach

This Bangladesh study used job-posting data, NLP preprocessing/tagging, and ARIMA to predict skills in demand. ([DU Journal of Business Studies](#))

Then your project says:

Existing studies demonstrate the usefulness of online job-posting data and NLP for analyzing skill demand in Bangladesh. However, our study uses more recent technology-job data and integrates job-role classification, salary analysis, and demand forecasting within a single ML pipeline.

Be careful with the last sentence: **verify the literature systematically before claiming this is a research gap.**

8. Add recent Bangladesh reports as supporting evidence

Don't use research papers alone.

For example, NSDA maintains Bangladesh skills-gap and labor-market studies, including an **Analysis of Skills Gap in the ICT Sector in Bangladesh**. ([National Security Directorate](#))

And CPD's 2025–2026 future-of-work research examines how AI, automation, digitalization and other forces may affect Bangladesh's labor market toward 2035. ([Centre for Policy Dialogue \(CPD\)](#))

These are useful for your:

Introduction → Related Work → Discussion → Future Work

But don't mix their projections directly with your job-posting data as though they measure the same thing.

9. Your complete ML pipeline

Now we can map your project **exactly** to the professor's rubric.

Stage 1 — Problem Formulation

Define:

Primary ML problem

Multi-class classification

| Predict the technology job category from job-posting information.

Secondary problem

Regression, only if salary data is sufficient:

| Predict advertised salary.

Additional analysis

Time-series forecasting

| Forecast demand trends for selected technology occupations.

This gives you:

```
Classification
      +
Regression
      +
Time Series
      +
NLP
```

That's a very strong academic project—but don't let the number of models become unmanageable.

10. Stage 2 — EDA

You should demonstrate:

Dataset overview

```
Number of rows
Number of columns
Data types
Missing values
Duplicates
```

Job analysis

```
Top job categories
Top job titles
Jobs by location
Jobs by experience
Jobs by month/year
```

Skill analysis

```
Top 20 skills
Skill frequency by role
Skill trend over time
```

Salary analysis

```
Salary distribution
Salary by experience
Salary by role
Salary by location
```

Correlation

For numerical features:

```
experience
salary
skill_count
job_age
etc.
```

This directly satisfies your professor's EDA requirement.

11. Stage 3 — Data preprocessing

Your pipeline could look like:

```
Raw Data
  ↓
Remove duplicates
  ↓
Handle missing values
  ↓
Normalize job titles
  ↓
Normalize locations
  ↓
Normalize salary
  ↓
Extract experience
  ↓
Extract skills using NLP
  ↓
Encode categorical features
  ↓
TF-IDF / embeddings for text
  ↓
Feature matrix
```

For example:

```
"React.js + Node.js + MongoDB developer"

  ↓

skills = {
  React,
  Node.js,
  MongoDB
}
```

12. Stage 4 — Feature engineering

This is where you can demonstrate real ML knowledge.

Create:

```
experience_years  
skill_count  
has_python  
has_java  
has_javascript  
has_sql  
has_react  
has_docker  
has_aws  
...
```

And possibly:

```
location_group  
salary_midpoint  
education_level  
experience_level  
year  
month
```

For text:

```
TF-IDF(job_description)
```

or, depending on your resources:

```
Sentence embeddings
```

For a course project, **TF-IDF is much easier to explain and reproduce.**

13. Stage 5 — Baseline model

Your professor explicitly requires a baseline.

For classification:

Baseline

Dummy Classifier

Then:

Logistic Regression

This gives you something meaningful to compare against.

14. Stage 6 — Advanced models

Use at least two:

Model A

Random Forest

Model B

XGBoost

Optionally:

SVM

So your experiment becomes:

Model	Type
Dummy Classifier	Baseline
Logistic Regression	Classical
Random Forest	Advanced
XGBoost	Advanced
SVM	Optional

Don't use 10 models just to make the project look bigger.

15. Stage 7 — Hyperparameter tuning

For example:

```
Random Forest
```

```
n_estimators  
max_depth  
min_samples_split  
min_samples_leaf
```

XGBoost:

```
n_estimators  
max_depth  
learning_rate  
subsample  
colsample_bytree
```

Use:

```
GridSearchCV
```

or preferably:

```
RandomizedSearchCV
```

if the parameter space is large.

16. Stage 8 — Cross-validation

For classification:

```
Stratified K-Fold  
K = 5
```

For example:

```
80% training
  ↓
5-fold CV
  ↓
Best parameters
  ↓
20% untouched test set
```

Do not perform preprocessing using the entire dataset before CV.

Use a pipeline:

```
Data
  ↓
Train fold
  ↓
Preprocessing
  ↓
Feature extraction
  ↓
Model
```

This prevents data leakage.

17. Time-series forecasting needs a different validation strategy

This is important enough to highlight.

Don't randomly shuffle your time-series data.

Instead:

```
Train: 2022
Test: 2023

Train: 2022-2023
Test: 2024

Train: 2022-2024
Test: 2025
```

This is closer to how forecasting actually works.

You can compare:

Baseline

Naive forecast

| Next month = previous month

Advanced

ARIMA / SARIMA / ETS

And potentially an ML model with lag features:

```
previous_month
previous_2_months
previous_3_months
rolling_mean
growth_rate
```

"Forecast technology job-demand trends and explore long-term future scenarios."

Why?

A 10-year numerical forecast from job advertisements has enormous uncertainty.

Interestingly, CPD's recent Bangladesh future-of-work study itself uses **foresight and scenario building** rather than claiming one deterministic future. ([Centre for Policy Dialogue \(CPD\)](#))

Your project can follow a similar philosophy:

```
Historical job data
  ↓
ML / Time-series forecasting
  ↓
Short-term quantitative forecast
  ↓
Long-term scenario discussion
```

+
External Bangladesh labor-market research

That is much more defensible academically.

19. Stage 9 — Error Analysis

This section can make your paper much stronger.

Don't just write:

| XGBoost achieved 91% accuracy.

Investigate **where it fails**.

For example:

Actual	Predicted
-----	-----
Data Analyst	Software Developer
DevOps Engineer	Software Developer
ML Engineer	Data Analyst

Then ask:

Why?

Maybe:

- Job titles overlap.
- Skills overlap.
- Companies use unusual titles.
- Descriptions are incomplete.
- Same job appears under different titles.

For salary regression:

Actual salary = 100k

Predicted = 65k

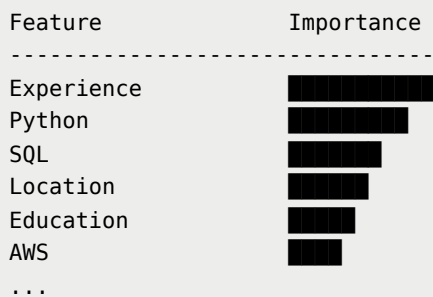
Investigate whether:

- Salary wasn't disclosed.
- Seniority was unclear.
- Location affected salary.
- Job category was unusual.

This directly satisfies your professor's **error analysis** requirement.

20. Stage 10 — Feature importance

For Random Forest/XGBoost you can show:



For a paper, you can discuss which characteristics contribute most to prediction.

Be careful to describe feature importance as **model behavior**, not automatically as causal influence.

21. Your final paper structure

You can almost directly map your work to the professor's required paper:

1. Abstract

Problem → Dataset → Methods → Results → Contribution

2. Introduction

```
Bangladesh tech employment
↓
Changing skill requirements
↓
Problem with traditional/static information
↓
Job portals provide dynamic data
↓
Research problem
↓
Our contribution
```

3. Related Work

Include:

- Bangladesh ICT job-posting study
- Bangladesh job-skill forecasting study
- Recent Bangladesh skills-gap reports
- Future-of-work research
- International job-market forecasting research

4. Dataset & EDA

Explain:

- Source
- Collection period
- Number of postings
- Features
- Missing values
- Cleaning
- EDA

5. Methodology

```
Data collection
↓
Cleaning
↓
NLP
↓
Feature engineering
↓
Train/Test
↓
Baseline
↓
Advanced models
↓
Hyperparameter tuning
↓
Evaluation
```

6. Experiments & Results

Table:

Model	Accuracy	Precision	Recall	F1
Dummy				
Logistic Regression				
Random Forest				
XGBoost				

Then charts.

7. Discussion & Error Analysis

Explain:

- Best-performing model
- Why it performed well
- Failure cases
- Feature importance
- Dataset limitations

8. Conclusion & Future Work

Mention:

- What you discovered
 - What model achieved
 - Practical usefulness
 - Limitations
 - Future expansion
-

22. The most important thing: your research contribution

your contribution around **integration + recent data + ML evaluation**.

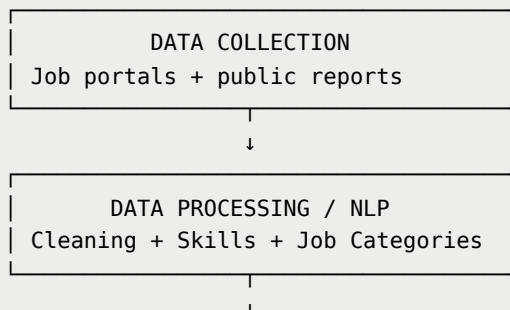
For example:

This study develops an end-to-end machine-learning framework for analyzing recent Bangladesh technology job postings, combining job-role classification, skill-demand analysis, salary exploration, and time-aware job-demand forecasting.

That's a much safer and stronger claim.

23. Recommended project architecture

I would organize your work into **4 layers**:





24. Your actual workflow

Phase 1 — Literature review

Collect around **8–15 good papers/reports**.

Categorize them:

- A. Job-market analysis
- B. Skill extraction
- C. Salary prediction
- D. Job-demand forecasting
- E. Bangladesh labor-market research
- F. Future-of-work research

Phase 2 — Dataset investigation

Before choosing your final ML problem, determine:

- What recent datasets actually exist?
- How many records?
- What years?
- Salary available?
- Job description available?

- Experience available?
- Location available?
- Can you legally/reliably collect the data?
- Can you get enough historical observations?

This phase determines your final methodology.

Phase 3 — Data collection

Build your dataset.

Target: **10k+ usable postings**, if realistically possible.

Phase 4 — Data cleaning + EDA

Phase 5 — NLP + feature engineering

Phase 6 — ML experiments

Baseline → advanced models → tuning → CV.

Phase 7 — Forecasting

Phase 8 — Error analysis

Phase 9 — Paper + presentation
